Panel data
$b_s, \beta, L = \beta b_s$
lay-ups (ABD)
end conditions
load n_0

Periodic cell (two pitches)

s_L s_R

Discretization (Section 2.1.3)

92 nodes, 93 segments

$w(\eta) = H_1 w_q + b_p H_2 \theta_q + H_3 w_{q'} + b_p H_4 \theta_{q'}$

GBT mode families

(side edges still free)

G global, D distortional, L local,
T transverse extension,
S shear, M Mindlin

Periodic modes: every mode takes equal values at s_L and s_R (Section 2.2)

In-plane modes
G, D, L, T
(each family)

$\mathbf{R}\,\mathbf{v} = \mathbf{0}$?
(Eq. (12))

Yes

keep $\mathbf{v}$

No

combine within the family:
$\mathbf{V}'_g \mathbf{Z}_g$ with $\mathbf{R}\mathbf{V}'_g \mathbf{Z}_g = \mathbf{0}$
missing periodic fields → T or D

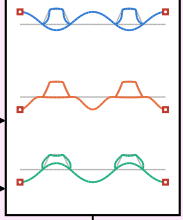
Warping u_k
of G and D

least shear strain energy, cell closed through s_L, s_R:
$\min\ \Pi_w(\mathbf{u})$ with $u_{s_L} = u_{s_R}$, $\mathbf{c}_{ax}^T \mathbf{u} = 0$ (Eqs. (9), (13))

Shear S, Mindlin M

merge the side-node functions: $u = u_{s_L} + u_{s_R}$, $\Omega = \Omega_{s_L} + \Omega_{s_R}$ (Eq. (14))

periodic basis
569 modes



Modal stiffness tensors

$\Xi_{de} = \int_b \mathbf{E}_d^T \mathbf{C}_{lam} \mathbf{E}_e \, ds$ (Eq. (26))

Pre-buckling state

$\delta U + \delta \Pi_{ext} = 0 \ \rightarrow\ S_{xx}, S_{ss}, S_{xs}$

Geometric tensors

$\mathbf{X}_{de}$ from S_{xx}, S_{ss}, S_{xs} (Eq. (29))

Ritz solution (cubic B-splines)

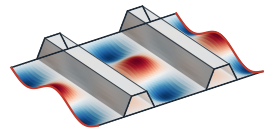
$(\mathbf{K} + \lambda \mathbf{G})\,\mathbf{a} = \mathbf{0}$, smallest $\lambda > 0$

other length or
end condition?

No

Yes (cross-section reused)

N_{cr}, buckling mode, ρ_g



example: critical mode of the Unsym cell, N_{cr} = 32.26 N/mm